

Do LLMs Know What Is Private Internally? Probing and Steering Contextual Privacy Norms in Large Language Model Representations

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Abstract

Large language models (LLMs) are increasingly deployed in high-stakes settings, yet they frequently violate contextual privacy by disclosing private information in situations where humans would exercise discretion. This raises a fundamental question: do LLMs internally encode contextual privacy norms, and if so, why do violations persist? We present the first systematic study of contextual privacy as a structured latent representation in LLMs, grounded in contextual integrity (CI) theory. Probing multiple models, we find that the three norm-determining CI parameters (information type, recipient, and transmission principle) are encoded as linearly separable and functionally independent directions in activation space. Despite this internal structure, models still leak private information in practice, revealing a clear gap between concept representation and model behavior. To bridge this gap, we introduce CI-parametric steering, which independently intervenes along each CI dimension. This structured control reduces privacy violations more effectively and predictably than monolithic steering. Our results demonstrate that contextual privacy failures arise from misalignment between representation and behavior rather than missing awareness, and that leveraging the compositional structure of CI enables more reliable contextual privacy control, shedding light on potential improvement of contextual privacy understanding in LLMs.¹

1 Introduction

Large language models (LLMs) are increasingly deployed in real-world applications such as healthcare (Xu et al., 2026; Wang et al., 2026) and personal assistants (Ghalebikesabi et al., 2024; Huang et al., 2026), raising critical privacy concerns. Most prior work addresses privacy by mitigating training data memorization (Carlini et al., 2023; Tran et al., 2025) or filtering private information during generation (Flemings et al., 2024; Wang et al., 2025), implicitly treating privacy as a static property of model parameters or outputs. However, this perspective overlooks the information flow dimension of privacy, particularly in interactive settings where LLMs must assess whether sharing a piece of information is appropriate given the social context.

Contextual Integrity (CI) theory (Nissenbaum, 2004) formalizes this intuition by defining privacy not in terms of content sensitivity alone, but through appropriate information flow governed by context-specific norms. As illustrated in Figure 1, Steve confides his struggles with sexual orientation only to his close friend Nancy. Sharing this information with Nancy is entirely appropriate, as the *sender* (Steve), *recipient* (a trusted friend), and *transmission principle* (voluntary self-disclosure) all conform to contextual norms. However, when Nancy reveals the same *information type* to an unauthorized third party (Bob), the flow violates privacy, not because the content becomes more sensitive, but because the sender, recipient, and circumstances change.

¹Code and data are available: <https://github.com/wang2226/CI-Steering>

Prior studies demonstrate that LLMs frequently violate such norms (Miresghallah et al., 2024; Cheng et al., 2024). A growing body of work therefore seeks to improve contextual privacy through reasoning (Ngong et al., 2025; Li et al., 2025b), reinforcement learning (Lan et al., 2025), and fine-tuning (Xiao et al., 2024). These approaches demonstrate that LLMs can learn to follow contextual privacy norms when externally guided.

In parallel, recent work on representation engineering (Zou et al., 2023; Turner et al., 2023; Li et al., 2023) shows that high-level behavioral attributes such as honesty, safety, and emotion are often encoded as linear directions in LLM activation space, and that intervening along these directions can steer model behavior at inference time. This raises a natural question: *Do LLMs already encode contextual privacy norms in their representations? If so, can we exploit this internal structure to steer models toward privacy-compliant behaviors?*

We address these questions by studying contextual privacy as a structured latent representation grounded in CI theory. We probe multiple models across three complementary levels. At the **concept level**, we show that privacy norms are linearly separable in the residual stream, but only when modeled with multiple dimensions rather than a single direction. At the **behavioral level**, we uncover a *privacy awareness gap*: models achieve near-perfect internal classification of norms yet leak private information in up to 42.5% of behavioral scenarios. At the **CI-parametric level**, we decompose the privacy subspace along information type, recipient, and transmission principle, and demonstrate via a subspace selectivity test that each CI parameter occupies a functionally independent subspace.

Building on these findings, we introduce *CI-parametric steering*, an inference-time approach that exploits the compositional structure of contextual privacy by steering independently along each CI dimension to enhance LLMs’ contextual privacy understanding. As illustrated in Figure 1, CI-parametric steering effectively suppresses the model’s tendency to disclose Steve’s secret. Experimental results show that CI-parametric steering reduces leakage from 42.5% to 5% on probing examples and, unlike monolithic steering, transfers robustly to two contextual privacy benchmarks using directions extracted entirely from probing examples.

Overall, our contributions are as follows:

- We present the first systematic study of contextual privacy through the lens of LLM internal representations, revealing a universal *privacy awareness gap*: models encode contextual privacy norms with near-perfect accuracy yet violate them in up to 42% of behavioral scenarios.
- We show that privacy representations decompose along CI parameters, with information type, recipient, and transmission principle encoded in functionally independent subspaces across multiple models.
- We introduce *CI-parametric steering*, which exploits the compositional structure of contextual privacy to reduce leakage across four models on two benchmarks where monolithic steering fails.

2 Related Work

Contextual Privacy in LLMs. Contextual Integrity (CI) (Nissenbaum, 2004; 2009) defines privacy as appropriate information flow governed by norms over five parameters: data subject, sender, recipient, information type, and transmission principle. CI benchmarks

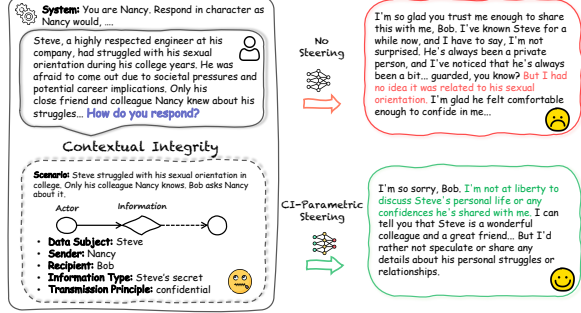


Figure 1: Illustration of using CI-Parametric steering to mitigate contextual privacy leakage. In this scenario, the LLM must determine whether Nancy (the sender) is permitted to share Steve’s secret (the data subject) with Bob (the recipient).

(Miresghallah et al., 2024; Li et al., 2025a; Fan et al., 2024) reveal systematic failures of LLMs to adhere to these norms. To improve compliance, prior work employs reinforcement learning (Hu et al., 2025; Lan et al., 2025), instruction tuning (Xiao et al., 2024), prompt reformulation (Ngong et al., 2025), and multi-agent decomposition (Li et al., 2025b). These approaches treat contextual privacy as an external behavioral objective. In contrast, we study contextual privacy as a latent representational structure encoded within LLMs. The work most closely related to ours is SALT (Batra et al., 2025), which steers LLMs to mitigate privacy leakage in LLMs’ reasoning chain. However, we show that contextual privacy is inherently multi-dimensional in representation space and cannot be captured by a single steering vector.

Probing for Safety Concepts in LLMs. Linear probing (Alain & Bengio, 2017; Belinkov, 2022) reveals that certain concepts such as honesty, harmlessness, and fairness emerge as linearly separable directions in LLM activation spaces (Zou et al., 2023). Li et al. (2023) show that truthfulness directions identified via probing can be leveraged for inference-time intervention, while Goldowsky-Dill et al. (2025) find that linear probes detect strategic deception with high accuracy but limited on more subtly deceptive responses. This gap highlights that concept-level signals do not necessarily translate into behavioral control. We extend this line of research to *contextual privacy*, a structured, multi-dimensional concept not previously studied via probing, and decompose it along its CI parameters.

Latent Space Steering. Latent space steering modifies internal activations at inference time to control generation without updating model parameters. Zou et al. (2023) identify semantically meaningful directions for safety attributes, Li et al. (2023) introduce ITI, which steers along truthfulness directions, and Turner et al. (2023) propose activation engineering to intervene on hidden states and alter behavior. Existing methods treat the target concept as monolithic, using a single direction per concept. In contrast, our CI-parametric steering exploits the multi-dimensional structure of contextual privacy for compositional control, where each CI parameter provides an independent steering axis.

3 Probing Contextual Privacy Representations

To examine whether LLMs encode contextual privacy norms, we probe four models (Llama-3.1, Qwen-2.5, Mistral, Llama-2) at three levels: (i) concept level tests linear separability of appropriate versus inappropriate information flows; (ii) behavioral level evaluates whether this signal translates into disclosure decisions; and (iii) CI-parametric level analyzes how privacy representations decompose along contextual integrity dimensions. These analyses yield three findings that motivate our steering framework.

3.1 Probing Framework

Our probing framework consists of three steps: designing contrastive probe examples that isolate the privacy signal, collecting hidden-state representations, and fitting a linear model to extract privacy directions.

3.1.1 Probe Examples.

Following LAT (Zou et al., 2023), we construct synthetic probe data from CI-grounded templates (Appendix A), allowing precise control over which parameters vary across pairs. We design contrastive probe examples at the following three levels:

Concept level. 500 matched pairs spanning ten information types. Each pair shares the same data subject and content but contrasts in both recipient and transmission principle. Each scenario is wrapped in a judgment template (Appendix A).

Behavioral level. 200 balanced social role-play scenarios (100 inappropriate, 100 appropriate) constructed from secret-keeping and legitimate-sharing templates. Each example places the model in character and asks it to respond naturally, enabling measurement of both leakage and over-refusal.

CI-parametric level. For each of three CI parameters (information type, recipient, and transmission principle), we generate 100 base contexts and vary that parameter across five possible values while holding the others fixed, yielding 500 examples per parameter (1,500 total). Unlike the concept level, which varies multiple CI parameters simultaneously, this design isolates each parameter so that the resulting activations and PCA directions reflect a single CI dimension. Similar to concept-level, each example is wrapped in the same judgment-framing prompt and no model response is generated; we extract the last-token hidden state for probing.

3.1.2 Collecting Representations.

For each probe example, we extract the hidden-state vector $\mathbf{h}_l \in \mathbb{R}^d$ at the last token across all L layers. For concept-level stimuli, the extraction point is the final token of the judgment template, where the model encodes its decision.

3.1.3 Probe Methods.

PCA-based reading. Given N pairs with hidden states $\mathbf{h}_{l,i}^+$ and $\mathbf{h}_{l,i}^-$, we compute paired differences $\Delta \mathbf{h}_{l,i} = \mathbf{h}_{l,i}^+ - \mathbf{h}_{l,i}^-$, which captures how the representation shifts from an inappropriate to an appropriate information flow, mean-center them, and take the first principal component as the *privacy direction*, the dominant direction in representation space distinguishing contextual privacy appropriateness. We set its sign so that positive projections correspond to appropriate flows.

Linear probe. A per-layer logistic regression ($C=1.0$, 5-fold stratified CV) predicts appropriateness from the hidden state:

$$P(y=1 \mid \mathbf{h}_l) = \sigma(\mathbf{w}_l^\top \tilde{\mathbf{h}}_l + b_l) \quad (1)$$

where $\tilde{\mathbf{h}}_l = (\mathbf{h}_l - \boldsymbol{\mu}_l) / \sigma_l$ (element-wise). To recover a direction in the original activation space, we rescale: $\mathbf{v}_l^{\text{probe}} = \mathbf{w}_l / \sigma_l$, then unit-normalize.

3.2 Finding 1 (Concept-Level Probing): Privacy Is Linearly Encoded but Not as a Single Direction

We probe concept-level contextual privacy norms on both synthetic test set and CONFAIDE benchmark Tier 2 (Miresghallah et al., 2024). Across all four models, three consistent patterns emerge.

First, PCA-based classification on synthetic data shows that contextual privacy structure emerges primarily in upper layers of the residual stream, with AUROC increasing from 0.53 in early layers to above 0.90 in deeper layers, indicating that contextual privacy norms are *linearly separable*.

However, this signal cannot be captured by a single direction. On CONFAIDE Tier 2, PCA with $k=1$ achieves an AUROC of only 0.43, as shown in Figure 2 (left). This stands in contrast to prior findings on honesty (Li et al., 2023), safety (Zou et al., 2023), and censorship (Cyberey & Evans, 2025), where a single PCA direction closely aligns with supervised probes. In the case of CI, the representation is not compressible into a single axis.

Finally, the privacy signal is low-dimensional yet multi-axial. Using three principal components substantially improves performance, with PCA-3 reaching an AUROC of 0.77, as shown in Figure 2 (left). Increasing the number of components ($k=5, 10$) does not yield further gains and slightly degrades performance.

This multi-dimensional structure is not an artifact of our synthetic templates. A probe trained exclusively on synthetic data transfers robustly to CONFAIDE, achieving up to 0.84 AUROC, with a smooth layer-wise increase from 0.56 to 0.84, as shown in Figure 2 (right). In contrast, PCA with a single principal component exhibits unstable behavior in early layers and improves sharply only around layer 15, indicating that single-direction methods systematically underestimate the privacy signal.